Physics-guided learning predicts battery thermal runaway from parsimonious features

Sihui Chen[1], Ryan Wang[2], Mengyuan Zhou[3, 4, 5], Shuguo Sun[3, 4, 5], Jun Xu[3, 4, 5*]

[1]*Department of Mechanical Engineering, University of Wisconsin-Madison, Madison, Wisconsin, 53706, USA*

[2]*Phillips Exeter Academy, 20 Main Street, Exeter, NH 03833, USA*

[3]*Energy Mechanics and Sustainability Laboratory (EMSLab), The Pennsylvania State University, University Park, PA 16802, USA*

[4]*Department of Mechanical Engineering, The Pennsylvania State University, University Park, PA 16802, USA*

[5]*Institute of Energy and the Environment, The Pennsylvania State University, University Park, PA 16802, USA*

[*]Corresponding author: Prof. Jun Xu, Email: junxu@psu.edu

Abstract: Thermal-runaway heat release determines the thermal-management and propagation-resistance requirements of battery systems, yet its calorimetric characterization is destructive, hazardous, and too time-consuming to cover the expanding diversity of cell designs and operating states. Existing predictive approaches often depend on specialized calorimetry, post-failure measurements, or high-dimensional inputs. Here, we develop a physics-guided feature-engineering framework to estimate capacity-normalized corrected heat release from routinely accessible cell data. An energy-balance description guides the construction of compact descriptors representing stored electrochemical energy and cell geometry. The framework is evaluated retrospectively using 338 fractional thermal-runaway calorimetry tests on 18650 and 21700 cylindrical cells. A composite descriptor formed from open-circuit voltage, state of charge, and capacity achieves a median mean absolute percentage error of 10.94% for 21700 cells, while two constructed descriptors achieve 13.84% for 18650 cells and 13.80% for the pooled dataset. These low-dimensional representations retain much of the predictive accuracy of a 16-variable baseline while requiring only routine pretest measurements. The results establish a compact surrogate for screening normalized heat-release behavior within the represented experimental domain. The methodology highlights the importance of physical laws in feature construction and selection and provides a basis for prioritizing experiments where calorimetry can deliver the greatest additional information.

Keywords: Battery safety; Thermal runaway calorimetry; Physics-guided machine learning; Feature engineering; Heat release

1 Introduction

The consequence of lithium-ion battery (LIB) thermal runaway (TR) is governed not by onset temperature alone, but by the energy delivered to the surroundings. Abuse can initiate internal short circuits and self-accelerating exothermic reactions [1], [2], while the resulting failure energy includes stored electrical energy, chemical reaction energy, and combustion of ejected material [3], [4]. Cell-to-cell propagation therefore depends on the total heat released, its rate, and partition between the casing and ejecta [5], [6]. These quantities vary with cell geometry, trigger method, internal electrochemical and thermal dynamics, and aging [7], [8], [9], with substantial scatter even among nominally identical cells. TR heat output is thus a variable engineering outcome that must be quantified directly.

Calorimetry provides the reference measurement. Copper-slug and bomb calorimetry quantified internal failure heat, combustion heat, and the respective contributions of electrical and chemical energy [3], [4]. Fractional thermal runaway calorimetry (FTRC) subsequently partitioned total heat output among the cell body, positive vent, and negative vent [5], [6], [7], [8]. The Battery Failure Databank now contains more than 300 FTRC tests across commercial cylindrical cells [10], providing an unprecedented basis for comparing heat release and failure variability. However, calorimetric qualification remains destructive, specialized, and replication-intensive. Masalkovaitė et al. reduced the required number of FTRC tests by predicting heat-output distributions from cell metadata and ejected mass [11]. Yet ejected mass is available only after TR; the method reduces calorimetry but does not remove the destructive experiment.

Existing prediction methods do not fully resolve this limitation. Mechanistic models describe internal-short-circuit heating, component-level reactions, electrochemical-thermal coupling, venting, and gas-phase processes [12], [13], [14], [15], [16], [17]. Physics-informed

neural networks and DeepONets further accelerate the prediction of temperature and reaction-state trajectories [18], [19], while recent data-amplification and inverse-learning methods improve thermochemical prediction and kinetic identification from limited experiments [20], [21]. These methods are valuable for mechanistic analysis, but generally require kinetic parameters, calibrated simulations, or completed abuse-test temperature histories. At the outcome level, continuous heat release has been predicted using post-failure mass [11], whereas pretest metadata have been used to classify a composite heat-and-ejection severity index [22]. Continuous, capacity-normalized total heat output has not been predicted from a compact and physically interpretable set of features available entirely before testing.

The missing link is feature engineering. Conventional workflows select from measured variables using statistical association [23] or algorithm-dependent importance [24], [25], [26]. Such methods can reduce dimensionality, but they neither guarantee physical relevance nor distinguish a readily available pretest variable from a costly post-test measurement. Our central premise is that governing equations should be used not only to simulate thermal runaway, but also to design the predictive feature space. As summarized, the proposed framework first maps available variables to the cell-level energy balance, then retains pretest variables connected to heat generation, thermal storage, external heating, or heat loss, and finally constructs composite descriptors that preserve the relationships implied by the governing physics. This approach introduces physics before model training, thereby improving interpretability, reducing dimensionality, and lowering experimental and computational costs (Fig. 1).

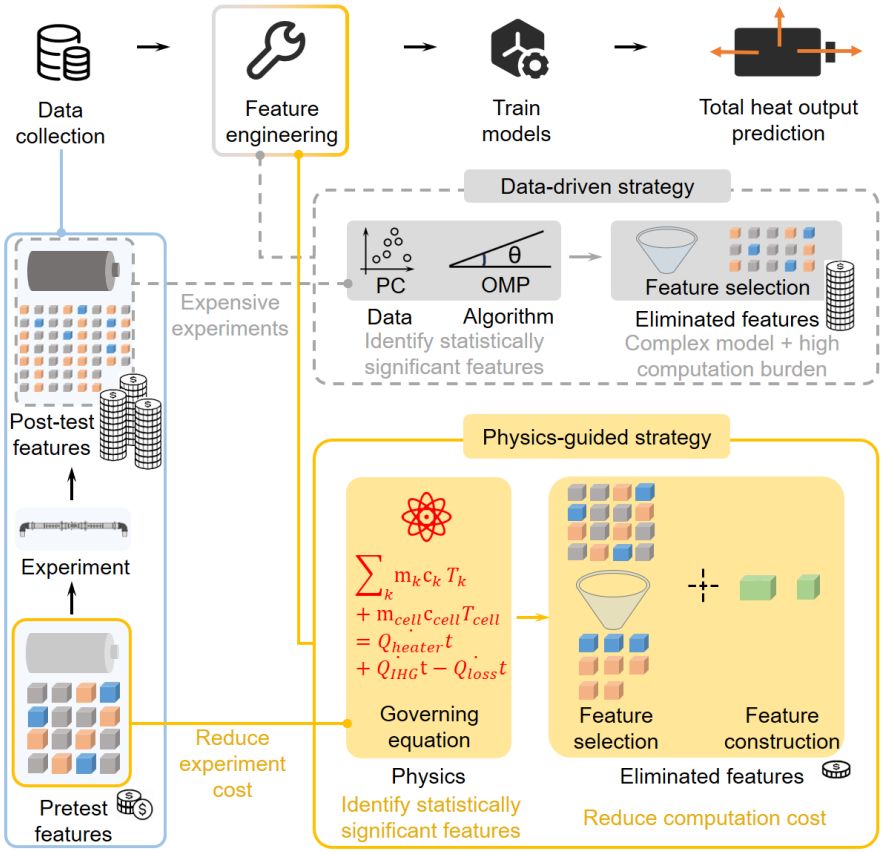


Fig. 1. Physics-guided pretest feature engineering for thermal-runaway heat-output prediction. Governing equations guide feature selection and construction, replacing dependence on costly post-test measurements with compact, physically interpretable pretest descriptors.

Here, we apply this framework to 338 FTRC tests of 18650 and 21700 cells from the Battery Failure Databank [10]. The governing energy balance transforms 16 recorded variables into descriptors representing stored electrochemical energy and geometry-normalized energy. Across 100 repeated random 80:20 train-test splits, one constructed descriptor produced a median mean absolute percentage error of 10.94% for 21700 cells; two descriptors produced errors of 13.84% for 18650 cells and 13.80% for the pooled dataset. These compact representations retained much of the accuracy of the full 16-variable baseline and, in several dimension-matched comparisons, outperformed models using post-test descriptors. The principal contribution is

therefore not another thermal-runaway model, but a physics-guided representation strategy that predicts continuous heat output using accessible information before failure. It provides a transparent basis for screening cells, prioritizing experiments, and directing calorimetric resources where they add the most information.

2 Physics-guided representation of thermal-runaway heat output

2.1 Thermodynamic basis of calorimetric heat output

Thermal runaway converts stored electrochemical energy into heat through coupled electrochemical, chemical, and thermal processes. This heat is subsequently redistributed among the cell, ejected material, calorimeter, and surroundings. Consequently, calorimetric heat output reflects both internal reaction energetics and heat transport across the experimental boundary. We examined these processes using an open dataset containing 338 fractional thermal runaway calorimetry (FTRC) tests [6]. The dataset comprises commercial 18650 and 21700 cylindrical cells from multiple manufacturers (Fig. S1 and Table S1). FTRC partitions the measured heat among the positive end, negative end, and cylindrical body of the cell [4]. Heat conducted through the casing or transported by ejected material is absorbed by adjacent calorimeter components. The absorbed heat is calculated from their temperature rise using the sensible-heat relationship $Q = cm\Delta T$ where Q, c, m, and ΔT denote absorbed heat, specific heat capacity, component mass, and temperature rise, respectively. This relationship defines the calorimetric measurement but does not describe the origin of the heat. We therefore introduced a global energy balance to connect the measured output with heat generation, storage, transfer, and loss [5]:

$$\sum_{k\in\{p,n\}} m_k c_k \Delta T_k + m_{cell} c_{cell} \Delta T_{cell} = \dot{Q}_{heater} t + \dot{Q}_{IHG} t - \dot{Q}_{loss} t \quad (1)$$

where m_k, c_k, and T_k denote the mass, specific heat capacity, and temperature rise of the kth calorimeter component, respectively. The subscript $k \in \{p,n\}$ refers to the copper meshes

positioned adjacent to the positive (p) and negative (n) ends of the cylindrical cell. m_{cell} , c_{cell} , and T_{cell} represent the mass, specific heat capacity, and temperature rise of the cell, respectively. $\dot{Q}_{heater}$ is the heater power, $\dot{Q}_{IHG}$ denotes the internal heat generation rate of the cell during thermal runaway, and $\dot{Q}_{loss}$ represents the heat loss rate due primarily to conduction, convection and radiation. t is the time interval considered during the thermal runaway test.

Eq. (1) separates the measurable thermal contribution from the internal electrochemical contribution. Thermal storage and external energy input are governed directly by cell mass, copper-mesh masses, heater power, and heating duration. Internal heat generation is more complex because it arises from coupled reactions and electrical failure processes [14], [15], [16], [17]. The internal heat generation term, Q_{IHG} , comprises the heat released from SEI decomposition (Q_{SEI}), anode-electrolyte side reactions (Q_a), electrical short-circuit or internal short-circuit (Q_{ec}), cathode-electrolyte side reactions (Q_c), and electrolyte decomposition (Q_e), thus:

$$Q_{IHG} = Q_{SEI} + Q_a + Q_{ec} + Q_c + Q_e \tag{2}$$

A complete calculation of Eq. (2) requires material-specific kinetic and thermodynamic parameters that are generally unavailable for commercial cells. However, mechanistic expressions still reveal which accessible variables govern the reaction-energy scale. For example, the internal-short-circuit contribution can be written in an Arrhenius-type form as [27],

$$\dot{Q}_{ec} = \frac{3600\eta VC}{V_{cell}} T_{ISC} A_{ec} SOC \exp\left(-\frac{Ea_{ec}}{RT}\right) \tag{3}$$

where η is the ISC factor; V , C , V_{cell} , and SOC denote the open-circuit voltage, cell capacity, cell volume, and state of charge, respectively, all of which can be obtained before the thermal runaway test. T_{ISC} is the ISC triggering temperature, A_{ec} is the pre-exponential factor, Ea_{ec} is the

activation energy associated with ISC, R is the universal gas constant ($8.314\ \mathrm{J\ mol^{-1}\ K^{-1}}$), and T is the cell temperature. Although these kinetic quantities cannot be resolved directly from the databank, Eq. (3) exposes four accessible dependencies, *i.e.*, voltage, capacity, SOC, and cell volume, that can be incorporated into the learned representation.

The remaining variables describe the triggering temperature, kinetic prefactor, activation energy, gas constant, and instantaneous cell temperature. These quantities are unavailable in the present dataset and cannot be used directly as predictive inputs. Eqs. (1)-(3) do not resolve the spatially and temporally evolving reaction network inside a failing cell. Instead, they provide an energy-conservation framework that identifies measurable variables connected to the integrated calorimetric response. Detailed derivations are provided in Supplementary Note 1.

2.2 Mapping the energy balance onto measurable variables

The governing equations contain substantially more information than the dataset can support. The key challenge is therefore to translate their physical structure into a representation based on quantities available before thermal runaway. To achieve this, we define a physics-informed feature as a measurable variable that appears explicitly in the governing equations or maps directly onto one of their terms. This definition is deliberately restrictive. A variable is retained because of its connection to energy generation, external forcing, thermal storage, or heat transfer, rather than its statistical association with heat output alone.

Within this framework, the available variables describe four aspects of the coupled cell-calorimeter system. First, open-circuit voltage (OCV), capacity, and state of charge (SOC) characterize the initial electrochemical state. OCV establishes the electrical potential, capacity determines the stored charge, and SOC specifies the accessible fraction of that charge. Together, these variables set the scale of the electrochemical energy available before failure and appear

explicitly in the ISC term of Eq. (3). Second, heater power and heater duration describe the external energy supplied to initiate thermal runaway. These quantities are not intrinsic cell properties; however, they establish the imposed thermal boundary condition of each FTRC test. Third, the masses of the cell and calorimeter components characterize thermal storage within the experimental system. Cell mass provides a measure of thermal inertia, whereas the positive and negative copper-mesh masses determine the sensible-heat capacities at the two cell ends. Finally, geometric quantities connect energy storage with heat transport. Cell volume defines the scale over which electrochemical energy is distributed, while surface area influences heat exchange with the calorimeter and surroundings. Geometry can therefore modify the measured heat output even when cells contain comparable amounts of stored energy.

We mapped each available variable onto its corresponding term in the governing equations, as summarized in Table. S2 (Fig. 2). The resulting primary feature space contained eight variables. The electrochemical subset comprised pretest OCV, capacity, and SOC, whereas the thermal subset comprised cell mass, positive and negative copper-mesh masses, heater power, and heater duration. This eight-variable representation spans both the energetic source and the calorimetric response. The electrochemical variables characterize the initial energy reservoir, while the thermal variables describe external forcing, heat storage, and calorimetric absorption. Their combination provides a compact representation of the coupled cell–calorimeter system without requiring unavailable chemistry-specific parameters.

The representation is necessarily coarse-grained. It does not explicitly describe cell chemistry, electrode architecture, degradation state, venting dynamics, or reaction kinetics. These unresolved factors may contribute to differences among cells with otherwise similar measured

properties. The objective is therefore not to reconstruct every microscopic process, but to constrain the feature space using the governing conservation law.

To determine how much predictive information is carried by each physical contribution, we constructed a hierarchy of comparative feature spaces. We evaluated the complete physics-informed set, the electrochemical and thermal subsets, and feature spaces from which these variables were removed. This design separates the contributions associated with the initial electrochemical state from those associated with the thermal boundary conditions. The complete feature-space definitions are provided in Table. S3, and their predictive performance is examined in *Sections 3.1* and *3.2*.

2.3 Construction of energy-based descriptors

Although the selected variables are individually connected to the governing physics, treating them as independent inputs does not preserve their functional relationships. A statistical model would otherwise need to infer these relationships from a relatively small and heterogeneous dataset. We therefore constructed three descriptors that explicitly encode the energy and geometric scales identified by the governing equations.

The first descriptor represents cell volume, which appears in Eq. (3) but is not directly reported in the dataset. We reconstructed it from the reported cell energy and volumetric energy density:

$$V_{cell} = \frac{E_{cell}}{\rho_{E,V}} \times 1000 \tag{4}$$

where E_{cell} is the nominal cell energy and $\rho_{E,V}$ is the volumetric energy density. This feature captures the geometric scale controlling volumetric energy release.

The accessible stored-energy descriptor was defined as

$$E_{store} = OCV \times C \times SOC \tag{5}$$

When OCV is expressed in volts and capacity in ampere-hours, E_{store} is expressed in watt-hours. It combines the cell's electrical state and accessible charge inventory into a single energy-scale variable. Normalizing this quantity by cell volume gives:

$$E_{store,norm} = \frac{OCV \times C \times SOC}{V_{cell}} \tag{6}$$

which represents the accessible electrochemical energy per unit cell volume. Among the constructed variables, $E_{\mathrm{store,norm}}$ most directly reflects the combined *OCV*, *C*, *SOC*, and V_{cell} dependence exposed by Eq. (3).

Together, the three descriptors represent complementary aspects of the initial cell state. V_{cell} captures the geometric scale, c captures the total electrochemical-energy scale, and $E_{\mathrm{store,norm}}$ captures volumetric energy loading. Their mathematical forms are prescribed by the governing relationships before model fitting. This approach differs from unconstrained feature generation because it embeds physical structure rather than relying solely on correlations among the original variables.

We evaluated these descriptors individually, in combination with one another, and together with the variables selected in *Section 2.2*. The complete feature-space definitions are provided in Tables. S4-S9, and their predictive performance is reported in *Section 3.3*. These comparisons test whether variation in thermal-runaway heat output can be represented using a small number of physically anchored energy scales. They do not assume that the descriptors resolve every underlying reaction or transport pathway. Instead, the descriptors provide a reduced-order representation consistent with global energy conservation and the information content of the available measurements.

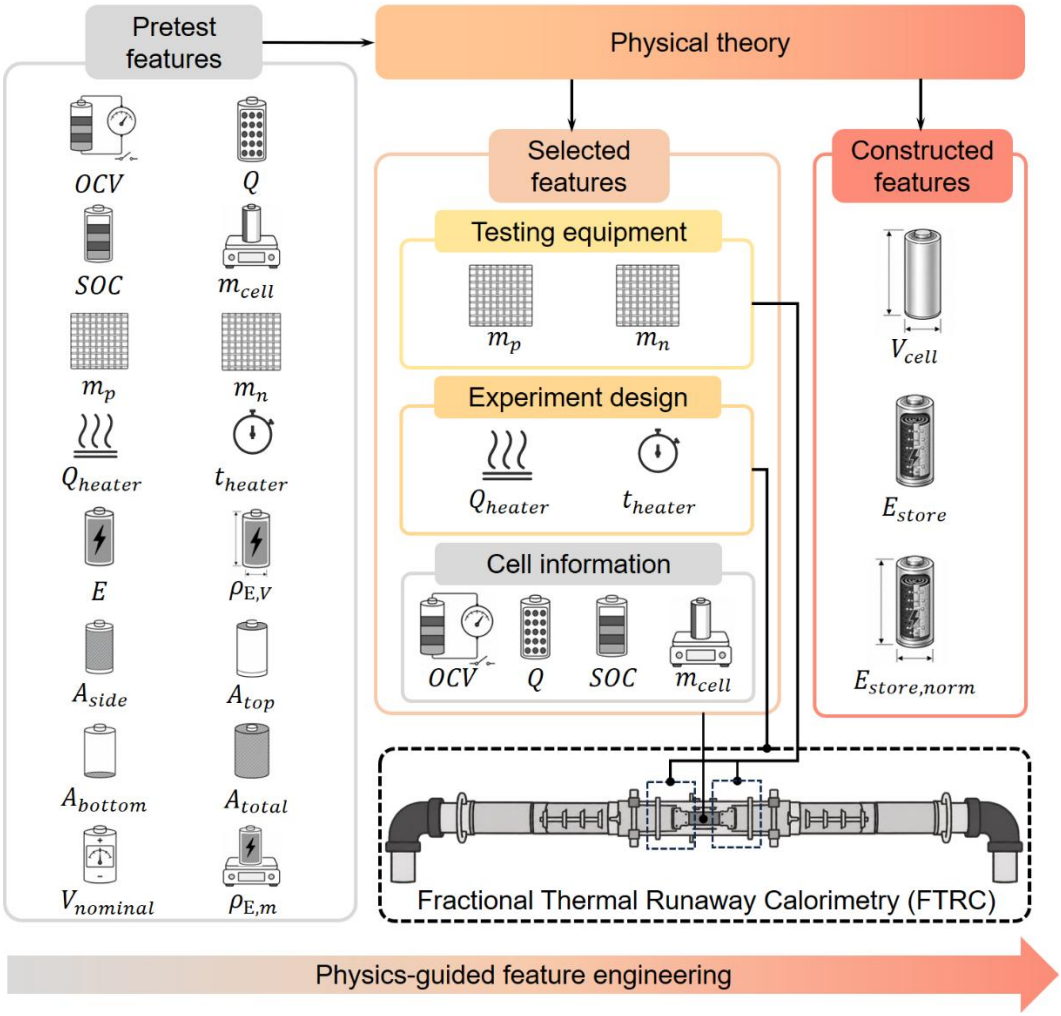


Fig. 2. Physics-guided selection and construction of pretest features for thermal-runaway heat-output prediction. Physical theory guides the selection of features related to testing equipment, experimental design, and cell information, and enables the construction of cell volume, stored electrochemical energy, and volume-normalized stored electrochemical energy.

3 Physics-guided descriptors concentrate predictive information in thermal-runaway heat output

3.1 Physical relevance preserves predictive information at low dimensionality

The energy balance developed in *Section 2* suggests that thermal-runaway heat output should depend more strongly on a small number of physically relevant quantities than on feature

number alone. We tested this hypothesis by comparing physics-informed representations with data-driven feature spaces of matched dimensionality. The prediction target was capacity-normalized corrected heat release, referred to below as heat output. Performance was evaluated using the median mean absolute percentage error (MAPE) across 100 random 80:20 training-testing splits. Complete preprocessing, model-selection, and validation procedures are described in the *Experimental Procedures*.

We first established how predictive performance changed when the feature space was reduced without physical guidance. Starting from the 16 available pretest variables, we randomly selected subsets containing 12, 9, 6, and 3 inputs. Both the median error and its variability generally increased as fewer variables were retained (Fig. 3(a)). The complete feature space therefore contained substantial predictive information, but arbitrary reduction did not reliably preserve that information. A different trend emerged when features were selected according to the governing energy balance. At matched dimensionality, the physics-informed subsets consistently produced lower errors and narrower error distributions than the randomly selected subsets (Figs. 3(b) and (c)). The difference was particularly pronounced in the low-dimensional regime. With only three inputs, the physics-informed representation achieved median MAPEs of 10.25%, 18.42%, and 17.33% for the 21700, 18650, and pooled datasets, respectively. The corresponding randomly selected subsets yielded substantially higher values of 14.30%, 40.69%, and 43.96%. These comparisons show that the loss of predictive accuracy associated with dimensionality reduction is not inevitable. A small feature space can retain much of the information contained in a larger one when its variables are connected to the physical origin and measurement of heat output. To further assess the effect of feature-selection strategy at matched dimensionality, we compared randomly selected, data-driven, and physics-guided six-feature subsets (Fig. 3(d)). The physics-guided

representation achieved median MAPEs of 8.41%, 13.61%, and 13.61% for the 21700, 18650, and pooled datasets, respectively. The data-driven subsets yielded corresponding values of 8.70%, 14.48%, and 14.58%; they were less accurate than the physics-guided subset for three datasets with different geometry size. The randomly selected subsets performed substantially worse, with median MAPEs of 10.90%, 18.30%, and 16.39%, respectively. Within the present dataset, the organization of the inputs was therefore more consequential than their number.

Predictive information must also be considered in relation to measurement accessibility. Six post-test variables generally produced lower errors than the same number of pretest variables (Fig. 3(e)), but they could be obtained only after thermal runaway had occurred. Under the cost assumptions summarized in Tables. S10 and S11, acquiring these variables was approximately ten times more expensive than acquiring the pretest information. Ejected mass illustrates this trade-off. It is closely associated with material expulsion and transported thermal energy, but its measurement still requires a complete destructive experiment.

Physics-informed pretest variables offer a different form of value. Although they do not directly observe the realized failure event, they are available before destructive testing and retain information connected to the governing energy balance. Their role is therefore not to replace thermal-runaway calorimetry, but to support the selection and prioritization of experiments. This distinction shifts the central question from how many variables can be collected to which measurements provide physically relevant information before testing begins.

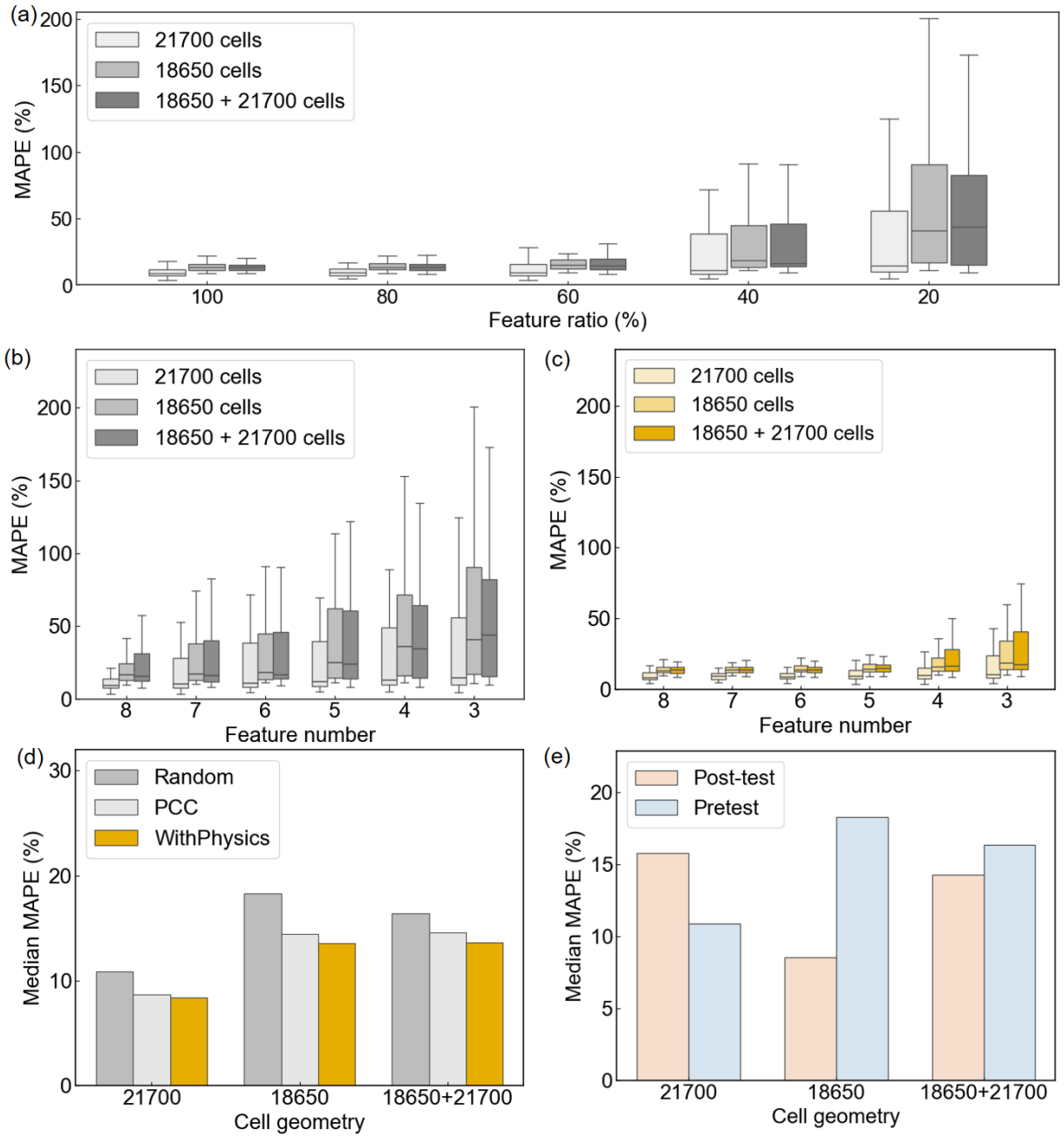


Fig. 3. Physical relevance preserves predictive information as feature dimensionality decreases. (a) Prediction errors obtained using different fractions of the 16 pretest variables. (b) Prediction errors of randomly selected pretest subsets containing three to eight variables. (c) Prediction errors of physics-informed subsets at matched dimensionality. (d) Comparison of feature spaces

constructed from randomly selected, data-driven, and physics-guided strategies. (e) Comparison of feature spaces constructed from pretest and post-test variables.

3.2 The initial electrochemical state defines the dominant energy scale

The preceding analysis established that physical selection preserves predictive information, but it did not reveal which part of the energy balance is most informative. We therefore separated the physics-informed variables into electrochemical and thermal groups. The electrochemical group contained OCV, capacity, and SOC, whereas the thermal group contained cell mass, positive and negative copper-mesh masses, heater power, and heater duration.

To reduce dependence on a specific regression architecture, we evaluated support vector machines [7], ElasticNet [8], [21], XGBoost [7], LightGBM [22], multilayer perceptrons (MLP) [28], and k-nearest neighbors (KNN) [29], [30]. The complete set of 16 pretest variables served as the data-driven baseline. We first examined whether the eight variables identified from the energy balance contained most of the information in the complete feature space. Despite using only half of the original inputs, the combined electrochemical and thermal set achieved performance comparable to the 16-variable baseline (Fig. S2). Conversely, removing these eight variables increased the median error to approximately six times the baseline value (Fig. S3). The predictive information in the original feature space was therefore concentrated in variables associated with energy storage, external heating, thermal inertia, and calorimetric absorption.

However, the electrochemical and thermal groups contributed differently to this information. The thermal variables alone produced median MAPEs of 14.61%, 47.88%, and 34.81% for the 21700, 18650, and pooled datasets, respectively (Figs. S4 and S5). These errors were substantially larger than those obtained using the complete feature space. Thermal mass and test conditions therefore did not provide a sufficient representation of heat output in the absence of

information about the cell's initial electrochemical state. The thermal variables nevertheless contributed differently across the three datasets. Their removal increased the median error for the 21700 cells but slightly reduced it for the 18650 and pooled datasets. This variation is consistent with a complementary, rather than dominant, role for thermal information. Geometry, thermal mass, and calorimetric coupling can modify how stored electrochemical energy appears as measured heat output, but their contribution depends on the range of cells and test conditions represented in the dataset. In contrast, the electrochemical variables retained considerably more predictive information. The electrochemical-only representation achieved median MAPEs of 10.69%, 12.64%, and 12.67% for the 21700, 18650, and pooled datasets, respectively (Fig. S6). Removing OCV, capacity, and SOC increased these values to 15.24%, 45.99%, and 33.14%. The corresponding error increases relative to the complete baseline were 74.67%, 249.52%, and 147.40% (Fig. S7). The strong deterioration after removing the electrochemical group identifies the initial energy state as the principal measured contribution to heat-output variation.

This result follows naturally from the physical representation developed in *Section 2*. OCV establishes the electrical potential, capacity determines the stored charge, and SOC specifies the accessible fraction of that charge. Together, these variables define the scale of electrochemical energy available before failure. Their predictive importance does not establish a unique reaction mechanism, but it is consistent with the energetic constraints imposed by the internal heat-generation term. Note that the relative contribution of the electrochemical group differed among datasets. The 18650 dataset contained 283 observations spanning 27 cell descriptions, whereas the 21700 dataset contained only 55 observations from two cell descriptions. Each description identifies a manufacturer, cell model, and any reported manual modification. The restricted size and diversity of the 21700 subset may limit the relationship that can be resolved between

electrochemical state and heat output. Under these conditions, thermal variables may provide additional information associated with geometry and heat transfer. In the pooled dataset, they may also help distinguish the two cylindrical form factors.

We therefore combined the electrochemical and thermal groups to determine whether they contained complementary information. The resulting eight-variable representation achieved median MAPEs of 8.62%, 13.11%, and 12.96% for the 21700, 18650, and pooled datasets, respectively (Fig. S8). These values matched or modestly improved upon the corresponding 16-variable baselines while using half as many inputs (Fig. S9).

Taken together, these comparisons establish a physical hierarchy within the available measurements. The initial electrochemical state defines the dominant energy scale, whereas thermal mass and boundary conditions refine how that energy appears in the calorimetric response. Their combination produced the most consistent performance across cell formats, although the magnitude of the thermal contribution remained sensitive to dataset composition (Fig. 4, Fig. S10).

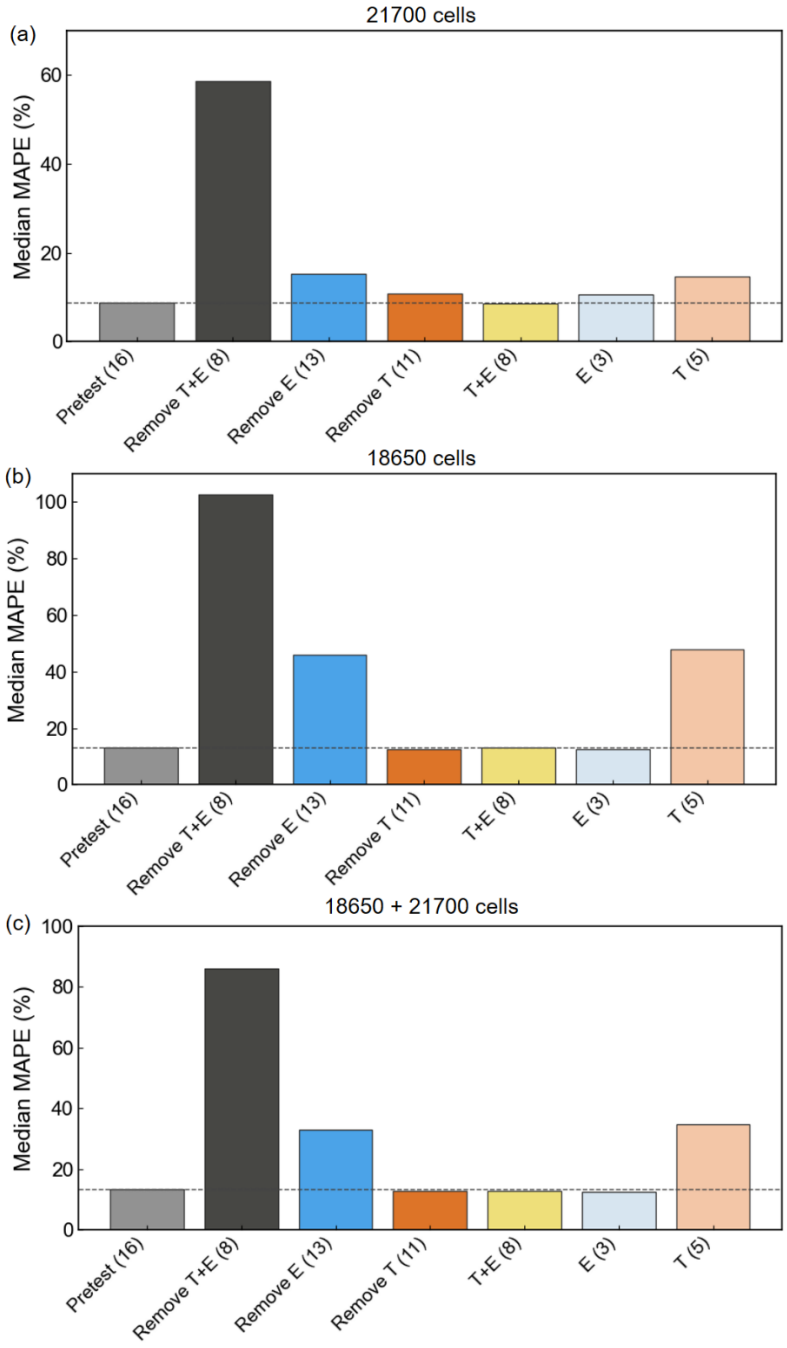


Fig. 4. Electrochemical and thermal contributions to heat-output prediction. Median MAPEs are compared for the complete data-driven baseline, the electrochemical and thermal subsets, their combination, and feature spaces excluding each physical group. Results are shown for (a) 21700 cells, (b) 18650 cells, and (c) the pooled dataset.

3.3 Energy-based descriptors compress the dominant physical information

The preceding analysis identified the initial electrochemical state as the principal source of predictive information. Yet OCV, capacity, and SOC entered the models independently, requiring their energetic relationship to be learned from limited data. We therefore asked whether the descriptors derived from the governing equations could encode this relationship directly and reduce the dimensionality of the representation.

We first introduced the constructed descriptors into the thermal feature space (Table. S4). Adding cell volume alone did not improve prediction, whereas adding E_{store} sharply reduced the median MAPE across all three datasets (Figs. S11 and S12). For the 21700, 18650, and pooled datasets, the error decreased from 14.61% to 9.12%, from 47.88% to 14.31%, and from 34.81% to 13.66%, respectively. These changes corresponded to relative reductions of 37.63%, 69.49%, and 60.49%. The contrast between cell volume and the energy-based descriptors distinguishes size from energetic state. Thermal variables describe external forcing, thermal inertia, and calorimetric absorption, while cell volume provides only a geometric scale. None directly quantifies the electrochemical energy available before failure. E_{store} supplies this missing information in a single descriptor, explaining its large improvement over the thermal-only representation. Indeed, the thermal variables combined with E_{store} has comparable performance as the eight-variable electrochemical-plus-thermal space for the all three datasets while requiring two fewer inputs (Figs. S13 and S14). By contrast, adding the constructed descriptors to the electrochemical feature space yielded smaller improvements (Figs. S15 and S16, Table. S6). This weaker response is expected because E_{store} is constructed from the same OCV, capacity, and SOC inputs. For the 21700 dataset, adding $E_{\mathrm{store,norm}}$ reduced the median MAPE from 10.69% to 10.42%. For the 18650 dataset,

adding V_{cell} and E_{store} reduced the median MAPE from 12.64% to 12.49%. For the pooled dataset, the same combination increased the error from 12.67% to 12.71%. Thees slightly variations suggest that the constructed features provide limited additional predictive value for electrochemical-only features.

We next removed the raw variables already embedded in the constructed descriptors (Tables. S7-8). The resulting representations retained or improved prediction despite their lower dimensionality (Figs. S17-S20). A two-descriptor representation has similar performance as the three-variable electrochemical space among three datasets. Thus, the constructed descriptors did more than supplement the raw inputs. They concentrated information distributed across several correlated variables. This conclusion was reinforced by models containing only the constructed descriptors. Cell volume or $E_{\mathrm{store,norm}}$ alone produced comparatively large errors, whereas space containing E_{store} approached the performance of the larger physics-informed spaces and the 16-variable baseline (Figs. S21 and S22). Cell size therefore carried limited predictive value without a corresponding energy scale. The dominant low-dimensional signal arose from the electrochemical energy accessible before failure.

Fig. 5 summarizes the strongest representation from each physical family (Fig. S23). For the 21700 cells, E_{store} alone achieved a median MAPE of 10.94%. Combining V_{cell} and E_{store} with the five thermal variables reduced the error to 9.12%, compared with 8.72% for the 16-variable baseline (Fig. 5(a)). For the 18650 cells, V_{cell} and E_{store} together achieved 13.84%. Their combination with the three electrochemical variables reached 12.49%, compared with 13.16% for the baseline (Fig. 5(b)). For the pooled dataset, the two descriptors achieved 13.80%, while the electrochemical variables combined with V_{cell} and E_{store} reached 12.71%, compared with 13.39%

for the baseline (Fig. 5(c)). These results establish two levels of physical information: (1) E_{store} captures the dominant energetic scale and is sufficient for a compact screening representation. Thermal variables then account for how this energy is coupled with the calorimetric system. Cell volume becomes useful when geometric variation is sufficiently broad, as in the pooled dataset; and (2) the best representation is therefore not universally the smallest one. Rather, the required dimensionality depends on whether the dataset varies primarily in electrochemical state or also in geometry and thermal boundary conditions.

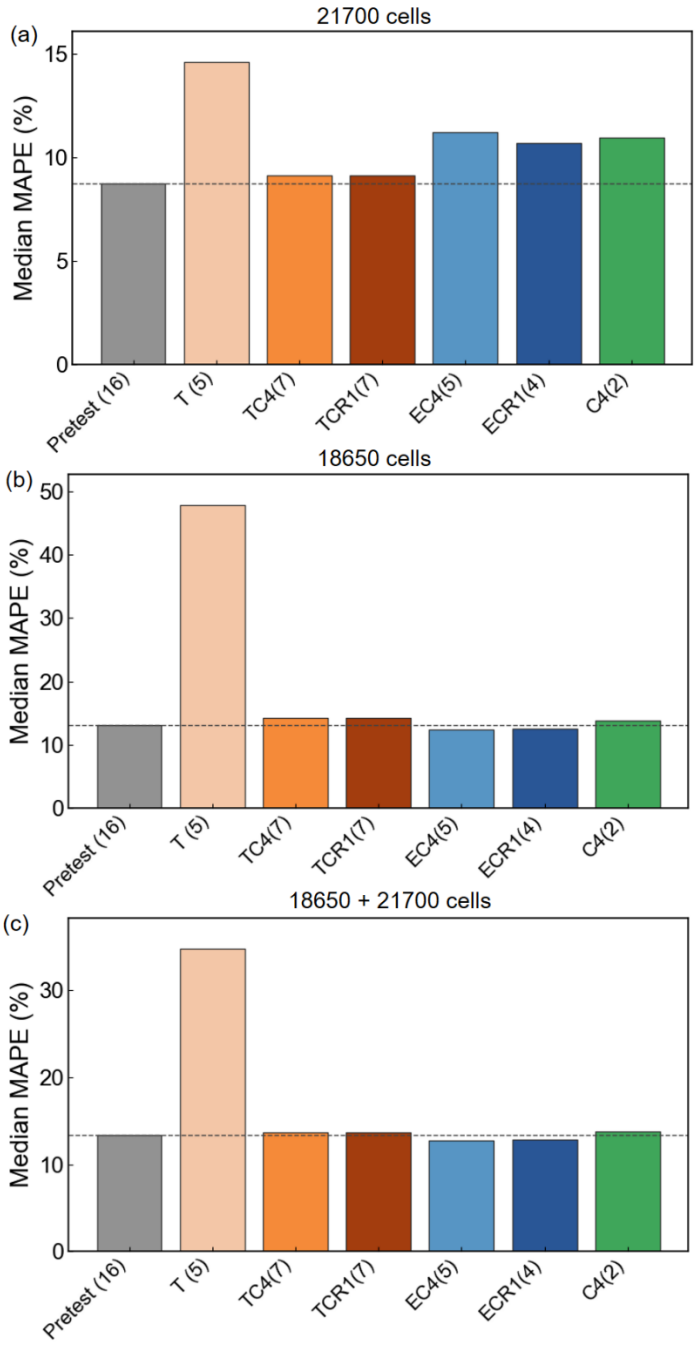


Fig. 5. Energy-based descriptors recover the predictive information of larger feature spaces. Median MAPEs of the 16-variable baseline, selected electrochemical and thermal spaces, and their combinations with constructed descriptors. Results are shown for (a) 21700 cells, (b) 18650 cells, and (c) the pooled dataset. Parentheses indicate the number of inputs.

Fig. 6 extends this comparison across the complete accuracy-dimensionality landscape. The constructed spaces occupy the lowest-dimensional region while remaining close to, or below, the 16-variable baseline (Fig. 6). For the 21700 cells, one constructed descriptor can achieve a relatively low error of 10.94%. For the 18650 cells, two descriptors retained most of the information present in the larger spaces. For the pooled dataset, two descriptors has similar prediction performance as the complete baseline. The physically constructed representations therefore shifted the usual trade-off between model accuracy and input dimensionality.

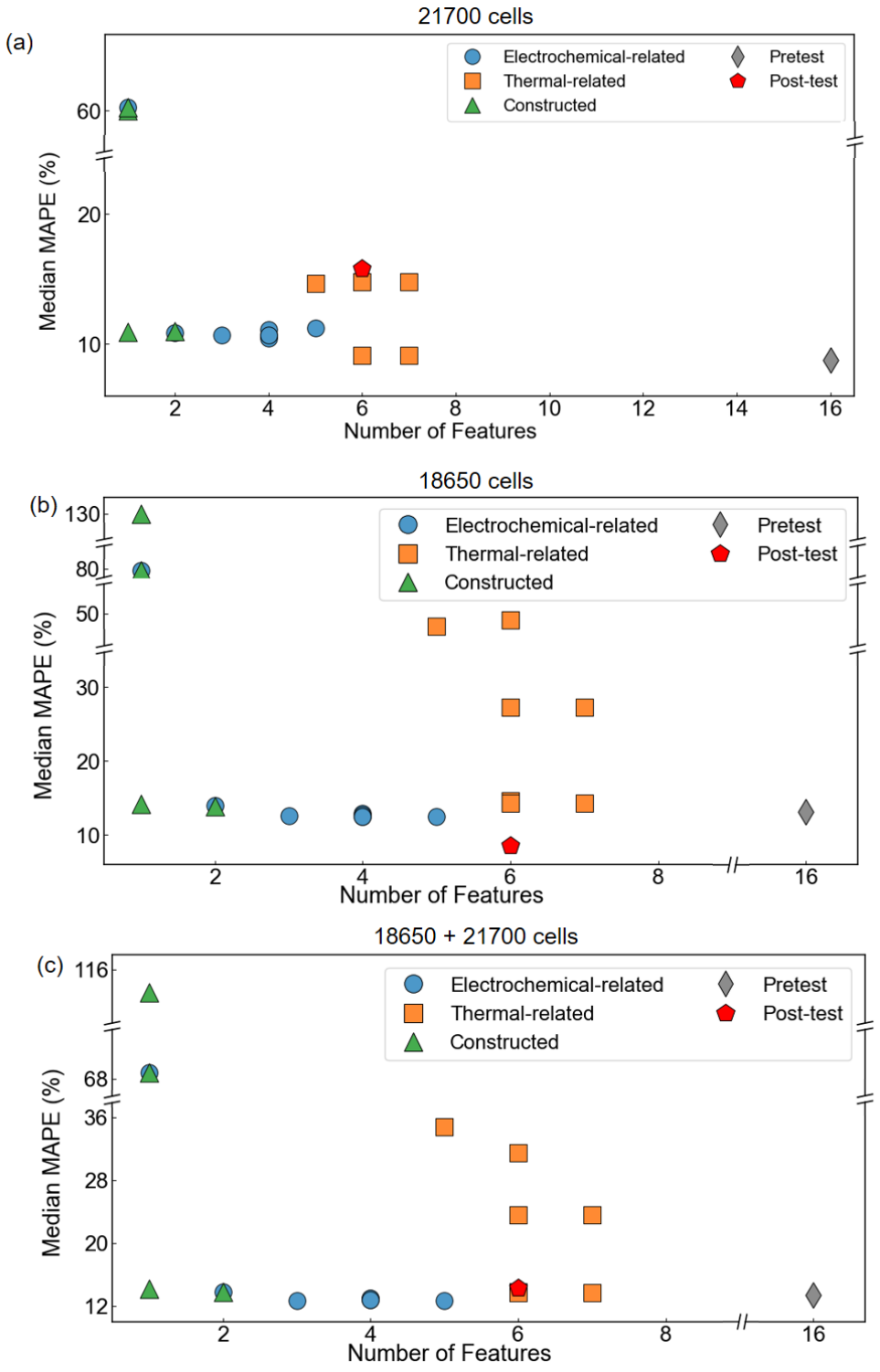


Fig. 6. Physical representation reshapes the accuracy-dimensionality relationship. Median MAPE as a function of input number for electrochemical, thermal, and constructed pretest

representations. The 16-variable data-driven baseline and six-variable post-test representation provide reference points. Results are shown for (a) 21700 cells, (b) 18650 cells, and (c) the pooled dataset.

The comparison with post-test variables further separates accessibility from information content. At matched dimensionality, physics-informed pretest spaces outperformed the corresponding post-test spaces (Figs. S24 and S25). Measurements collected after failure are not inherently more informative. Their value depends on whether they capture a governing energy scale that is distinct from information already available before testing.

Therefore, it is safe to conclude that the initial stored-energy scale contains most of the accessible pretest information about heat output. Thermal and geometric descriptors supply secondary information when cell format and calorimetric coupling introduce additional variation. This interpretation is physical but not mechanistic in a strict causal sense. The performance of E_{store} does not imply that stored electrical energy is converted completely into measured heat or that ISC is the sole heat-generation pathway. Instead, it identifies stored energy as an effective system-level coordinate for organizing heat-output variation within the present dataset.

4 From predictive parsimony to experimental efficiency

Predictive accuracy alone does not determine whether a model can improve thermal-runaway assessment. The required inputs must also be considered in terms of when they become available and what is required to obtain them. The present results reveal two distinct forms of efficiency: reduced measurement burden and reduced representational complexity (Fig. 7).

The first arises from replacing post-test measurements with information available before TR. Variables such as ejected mass may be highly informative, but their acquisition requires a completed destructive experiment. In contrast, OCV, capacity, SOC, and nominal cell

specifications can be obtained without triggering failure. Under the cost assumptions summarized in Tables. S10 and S11, acquiring the pretest information was approximately one order of magnitude less expensive than acquiring the post-test variables (Fig. 7(a)). This comparison should not be interpreted as a realized tenfold reduction in the total cost of a safety-testing program. Physics-guided estimation still depends on a calorimetry dataset for model development and requires continued experimental validation. The analysis instead quantifies the difference in feature-acquisition burden once a validated predictive framework has been established.

The second efficiency arises from physical compression of the input space. The complete data-driven baseline used 16 pretest variables, whereas the lowest-dimensional competitive representations required only one or two constructed descriptors (Fig. 7(b)). This reduction simplifies data collection, limits the number of required inputs, and improves the physical transparency of the model.

Together, these two forms of efficiency suggest a complementary role for physics-guided prediction in calorimetry planning. Within a validated domain, routine pretest information could provide an initial estimate of capacity-normalized corrected heat release. Destructive experiments could then remain focused on validation, previously unseen designs, and conditions for which existing data provide insufficient support.

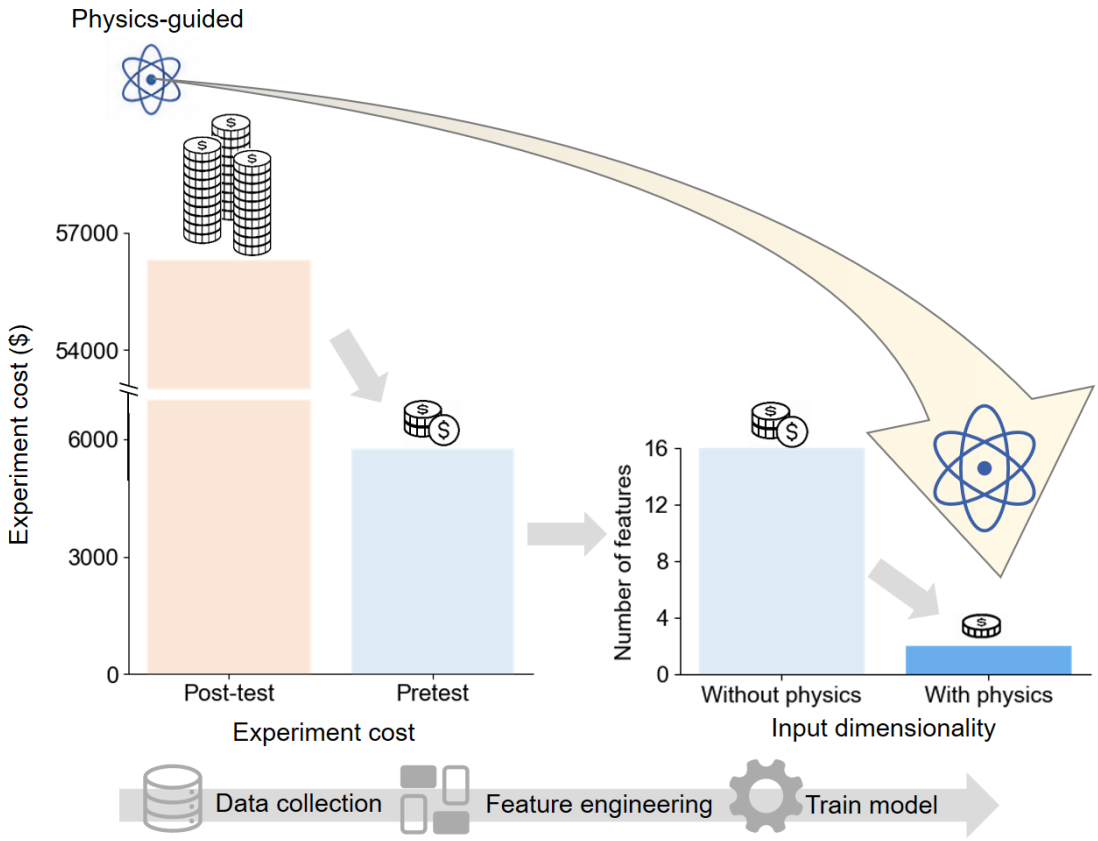


Fig. 7. Physics-guided representation reduces measurement burden and input dimensionality. (a) Estimated acquisition costs of pretest and post-test feature spaces under the cost assumptions reported in Tables. S10 and S11. (b) Comparison between the 16-variable data-driven baseline and the lowest-dimensional competitive physics-informed representation. Feature number describes input dimensionality rather than computational cost. Physics-guided estimation is intended to complement, not replace, destructive calorimetry.

5. Concluding Remarks

This study establishes a physics-guided framework for representing TR heat output using routine cell and test information. Beginning from the governing energy balance, the framework maps measurable variables onto relevant physical processes and preserves their relationships through energy-based descriptor construction.

Across the assembled FTRC dataset, one or two descriptors achieved median MAPEs of 10.94%-13.83% under repeated random splits. These compact representations retained much of the predictive information contained in the complete 16-variable pretest space. The result shows that physically organized inputs can be more informative than larger, unconstrained feature sets.

The stored-energy descriptor E_{store} was the common element of the lowest-dimensional competitive representations. Its prominence identifies the initial electrochemical-energy scale as the principal coordinate organizing heat-output variation within the investigated dataset. Thermal and geometric variables provided complementary information when cell format and calorimetric coupling introduced additional variation.

This interpretation does not imply complete conversion of stored electrical energy into measured heat or identify a unique reaction pathway. TR involves multiple exothermic reactions, material ejection, incomplete energy capture, and heat transfer beyond the calorimetric boundary. E_{store} is therefore a reduced system-level descriptor rather than a mechanistic measure of reaction enthalpy.

More broadly, this work shows how governing physics can guide learning when experimental data are limited and costly. Organizing accessible measurements around conserved quantities can reduce dimensionality while preserving information relevant to the target observable. For TR assessment, this principle provides a foundation for combining predictive screening with targeted calorimetry.

6. Experimental Procedures

6.1 Study design and data source

This study retrospectively analyzed the open-access Battery Failure Databank [6]. The analysis used 338 fractional thermal runaway calorimetry (FTRC) tests of cylindrical lithium-ion

cells. The dataset comprised 283 tests of 18650 cells across 27 cell descriptions and 55 tests of 21700 cells across two descriptions. Cell descriptions distinguish manufacturer, cell model, and reported cell modifications. Their distribution is provided in Table. S1.

The databank contains measurements collected using several FTRC generations and thermal-runaway trigger mechanisms. These include nail penetration, external heating, and heating with an internal-short-circuit device. The reported corrected total energy yield accounts for conductive heat loss and the energy associated with unrecovered cell mass [6]. Experimental procedures and calorimeter corrections were adopted from the original databank publication.

Three analysis cohorts were considered. Separate models were developed for the 18650 and 21700 cells to examine form-factor-specific behavior. A pooled cohort containing both form factors was used to assess prediction across a broader range of cell geometries and capacities.

6.2 Prediction target

The prediction target was the capacity-normalized corrected heat release, $y_i = \frac{Q_{\mathrm{corr},i}}{C_i}$, where $Q_{\mathrm{corr},i}$ is the corrected total energy yield of test *i*, expressed in kJ. C_i is the corresponding cell capacity in Ah. The resulting target has units of kJ/Ah. This normalization reduces the direct dependence of heat release on cell capacity and facilitates comparisons across cell formats.

Throughout the manuscript, this quantity is termed the capacity-normalized corrected heat release. The shorter-term heat output is used only after this definition and refers to the same normalized quantity.

6.3 Input variables and data preprocessing

The initial feature space contained 16 variables describing the cell state, cell geometry, calorimeter configuration, and thermal-runaway trigger conditions. These variables included OCV, capacity, SOC, nominal voltage, cell energy, volumetric energy density, specific energy, cell mass,

positive and negative copper-mesh masses, heater power, heater time, and four casing-area descriptors. Variable definitions, units, and source columns are reported in Table. S2.

Entries marked as unavailable in the source databank were parsed as missing values. Heater-related fields that were inapplicable to nail-triggered experiments were treated as missing rather than as zero. Incomplete records were retained because removing them would substantially reduce the available sample size.

Within each repeated analysis, missing values were replaced by the mean of the corresponding feature in the training partition. The resulting training means were then applied to the held-out test partition. This ordering prevents information from the test data from entering the imputation procedure. Features requiring scale normalization were transformed using parameters estimated exclusively from the training partition.

SOC is not provided as a dedicated column in the archived databank workbook. Its derivation from the reported test conditions must therefore be specified explicitly, including any conversion, assignment, or lookup procedure. The derivations of cell volume, surface areas, volumetric energy density, and specific energy must likewise be reported with their units.

6.4 Physics-guided feature representations

Candidate variables were organized according to the energy balance developed in *Section 2*. This mapping connected each measurable input to internal heat generation, external heating, thermal storage, or heat transfer. It was defined before model fitting and did not use the observed prediction errors.

The electrochemical feature group, denoted E, contained OCV, capacity, and SOC. These variables describe the cell's initial electrical state and provide information about the electrochemical energy available before failure. The thermal feature group, denoted T, contained

cell mass, positive and negative copper-mesh masses, heater power, and heater time. These variables represent cell thermal inertia, calorimeter thermal storage, and the externally supplied trigger energy. Heater time requires particular care in interpreting feature availability. It is determined only after heating begins and therefore is not a strictly pretest cell descriptor. Feature spaces containing heater time are consequently described as using pretest and trigger-protocol information. Only feature spaces constructed entirely from cell information available before heating are classified as strictly pretest representations. Three additional descriptors were constructed from the governing relationships (*Section 2.3*, Table S4).

The complete feature-space design is reported in Tables S3-S9. The comparisons included the full 16-variable baseline, the E and T groups, their combination, constructed-only representations, and combinations of selected and constructed descriptors. Redundant raw variables were removed in designated feature spaces to determine whether the constructed descriptors could replace their constituent measurements.

For the feature-number baseline, target-independent subsets containing 12, 9, 6, and 3 of the 16 variables were generated. These subsets represented approximately 80%, 60%, 40%, and 20% of the original feature space.

6.5 Model development and hyperparameter selection

Six regression algorithms were evaluated to reduce dependence on a single model class: support vector machine (SVM), ElasticNet, XGBoost, LightGBM, multilayer perceptron (MLP), and k-nearest-neighbor (KNN) [7], [8], [21], [22], [28], [29], [30]. These algorithms span kernel-based, regularized linear, tree-ensemble, neural-network, and local-neighborhood approaches.

Model development followed a repeated nested evaluation structure. For each repetition, 80% of the records were assigned to the training partition and 20% to the held-out test partition.

Preprocessing, hyperparameter optimization, and model selection were confined to the training data. Hyperparameters were selected using cross-validation within that partition. The selected pipeline was then fitted to the complete training partition and evaluated once on the held-out test data.

Within each training partition, model hyperparameters were optimized through grid search method using shuffled 5-fold cross-validation. Early stopping was applied to XGBoost, LightGBM, and MLP, while the remaining algorithms were trained without early stopping. The complete hyperparameter search spaces are provided in Table. S12. All analyses were conducted using Python 3.14.

The primary evaluation used random splitting at the individual-test level. Consequently, tests sharing a cell description or experimental series could appear in both partitions. The resulting errors therefore quantify interpolation within the sampled databank domain. They do not measure generalization to unseen cell designs, manufacturers, calorimeter generations, or experimental campaigns.

For each feature space, algorithm selection must be based only on inner cross-validation performance. Selecting the best algorithm from held-out test errors would introduce optimistic bias.

6.6 Performance evaluation

The train-test procedure was repeated 100 times using different random partitions. Predictive error in each held-out partition was quantified using mean absolute percentage error (MAPE),

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\% \quad (1)$$

where n is the number of data samples, while y_i and $\hat{y}_i$ are the real and predicted values of data sample i.

Model performance was summarized by the median MAPE across the 100 repetitions. The median was selected because occasional unfavorable partitions produced strongly right-skewed error distributions. The interquartile range should accompany the median wherever the distribution across repetitions is shown.

Because the repeated test partitions overlap, the 100 MAPE values are not independent experimental replicates. Their distribution is therefore interpreted descriptively and is not used for conventional significance testing. Comparisons among feature spaces quantify robustness to random sampling within the available dataset.

6.7 Estimation of feature-acquisition cost

The feature-acquisition analysis compared the resources required for pretest characterization with those required for FTRC and post-test measurements. Costs included cell procurement, measurement equipment, experimental labor, and post-test data processing. Labor cost was calculated from the reported task duration, personnel requirement, and hourly compensation rate. The underlying assumptions and source prices are reported in Tables. S10 and S11.

All costs are reported in US dollars for a defined price year. Equipment costs are distinguished from per-test operating costs because capital equipment can be reused across experiments. The analysis represents an order-of-magnitude comparison rather than a complete techno-economic assessment, unless depreciation, facility overhead, safety infrastructure, consumables, and equipment utilization are included.

6.8 Data and code availability

The Battery Failure Databank is publicly available from the National Renewable Energy Laboratory at https://www.nrel.gov/transportation/battery-failure.html [6].

Declaration of competing interests

The authors declare that they have no known competing financial interests that could have appeared to influence the work reported in this paper.

Acknowledgements

JX acknowledges financial support from Pennsylvania State University through faculty startup funds.

Supporting information

Supplemental information includes Supplementary Note 1, Figures S1-S25, and Tables S1-S11 and is available online with this article.

References

[1] X. Feng, M. Ouyang, X. Liu, L. Lu, Y. Xia, and X. He, "Thermal runaway mechanism of lithium ion battery for electric vehicles: A review," *Energy Storage Mater.*, vol. 10, pp. 246–267, Jan. 2018, doi: 10.1016/j.ensm.2017.05.013.

[2] Q. Wang, B. Mao, S. I. Stoliarov, and J. Sun, "A review of lithium ion battery failure mechanisms and fire prevention strategies," *Prog. Energy Combust. Sci.*, vol. 73, pp. 95–131, Jul. 2019, doi: 10.1016/j.pecs.2019.03.002.

[3] X. Liu, S. I. Stoliarov, M. Denlinger, A. Masias, and K. Snyder, "Comprehensive calorimetry of the thermally-induced failure of a lithium ion battery," *J. Power Sources*, vol. 280, pp. 516–525, Apr. 2015, doi: 10.1016/j.jpowsour.2015.01.125.

[4] R. E. Lyon and R. N. Walters, "Energetics of lithium ion battery failure," *J. Hazard. Mater.*, vol. 318, pp. 164–172, Nov. 2016, doi: 10.1016/j.jhazmat.2016.06.047.

[5] W. Q. Walker *et al.*, "Decoupling of heat generated from ejected and non-ejected contents of 18650-format lithium-ion cells using statistical methods," *J. Power Sources*, vol. 415, pp. 207–218, Mar. 2019, doi: 10.1016/j.jpowsour.2018.10.099.

[6] D. P. Finegan *et al.*, "Modelling and experiments to identify high-risk failure scenarios for testing the safety of lithium-ion cells," *J. Power Sources*, vol. 417, pp. 29–41, Mar. 2019, doi: 10.1016/j.jpowsour.2019.01.077.

[7] W. Q. Walker *et al.*, "The effect of cell geometry and trigger method on the risks associated with thermal runaway of lithium-ion batteries," *J. Power Sources*, vol. 524, p. 230645, Mar. 2022, doi: 10.1016/j.jpowsour.2021.230645.

[8] M. Sharp *et al.*, “Thermal Runaway of Li-Ion Cells: How Internal Dynamics, Mass Ejection, and Heat Vary with Cell Geometry and Abuse Type,” *J. Electrochem. Soc.*, vol. 169, no. 2, p. 020526, Feb. 2022, doi: 10.1149/1945-7111/ac4fef.
[9] S. Sun *et al.*, “Aging matters: How degradation pathways govern thermal runaway in lithium-ion batteries,” *J. Energy Chem.*, vol. 114, pp. 10–21, Mar. 2026, doi: 10.1016/j.jechem.2025.09.093.
[10] D. P. Finegan *et al.*, “The battery failure databank: Insights from an open-access database of thermal runaway behaviors of Li-ion cells and a resource for benchmarking risks,” *J. Power Sources*, vol. 597, p. 234106, Mar. 2024, doi: 10.1016/j.jpowsour.2024.234106.
[11] K. Masalkovaitė, P. Gasper, and D. P. Finegan, “Predicting the heat release variability of Li-ion cells under thermal runaway with few or no calorimetry data,” *Nat. Commun.*, vol. 15, no. 1, p. 8399, Sep. 2024, doi: 10.1038/s41467-024-52653-3.
[12] J. Deng, C. Bae, J. Marcicki, A. Masias, and T. Miller, “Safety modelling and testing of lithium-ion batteries in electrified vehicles,” *Nat. Energy*, vol. 3, no. 4, pp. 261–266, Apr. 2018, doi: 10.1038/s41560-018-0122-3.
[13] G.-H. Kim, A. Pesaran, and R. Spotnitz, “A three-dimensional thermal abuse model for lithium-ion cells,” *J. Power Sources*, vol. 170, no. 2, pp. 476–489, Jul. 2007, doi: 10.1016/j.jpowsour.2007.04.018.
[14] P. T. Coman, E. C. Darcy, C. T. Veje, and R. E. White, “Modelling Li-Ion Cell Thermal Runaway Triggered by an Internal Short Circuit Device Using an Efficiency Factor and Arrhenius Formulations,” *J. Electrochem. Soc.*, vol. 164, no. 4, p. A587, Jan. 2017, doi: 10.1149/2.0341704jes.
[15] D. Ren *et al.*, “Model-based thermal runaway prediction of lithium-ion batteries from kinetics analysis of cell components,” *Appl. Energy*, vol. 228, pp. 633–644, Oct. 2018, doi: 10.1016/j.apenergy.2018.06.126.
[16] X. Feng *et al.*, “A Coupled Electrochemical-Thermal Failure Model for Predicting the Thermal Runaway Behavior of Lithium-Ion Batteries,” *J. Electrochem. Soc.*, vol. 165, no. 16, p. A3748, Dec. 2018, doi: 10.1149/2.0311816jes.
[17] J. Kim, A. Mallarapu, D. P. Finegan, and S. Santhanagopalan, “Modeling cell venting and gas-phase reactions in 18650 lithium ion batteries during thermal runaway,” *J. Power Sources*, vol. 489, p. 229496, Mar. 2021, doi: 10.1016/j.jpowsour.2021.229496.
[18] S. W. Kim, E. Kwak, J.-H. Kim, K.-Y. Oh, and S. Lee, “Modeling and prediction of lithium-ion battery thermal runaway via multiphysics-informed neural network,” *J. Energy Storage*, vol. 60, p. 106654, Apr. 2023, doi: 10.1016/j.est.2023.106654.
[19] J. Jeong, E. Kwak, J. Kim, and K.-Y. Oh, “Prediction of thermal runaway for a lithium-ion battery through multiphysics-informed DeepONet with virtual data,” *eTransportation*, vol. 21, p. 100337, Sep. 2024, doi: 10.1016/j.etran.2024.100337.
[20] Y. Wang *et al.*, “Temperature excavation to boost machine learning battery thermochemical predictions,” *Joule*, vol. 8, no. 9, pp. 2639–2651, Sep. 2024, doi: 10.1016/j.joule.2024.07.002.
[21] G. Wang *et al.*, “Calorimetry-free identification of battery reaction kinetics using simulation-driven deep learning,” *Energy Storage Mater.*, vol. 88, p. 105087, May 2026, doi: 10.1016/j.ensm.2026.105087.
[22] Q. Hu, M. Wu, C. Wang, Y. Da, and H. Du, “Safety-oriented pre-event severity prediction of lithium-ion battery thermal runaway using interpretable machine learning,” *Process Saf. Environ. Prot.*, vol. 211, p. 108830, May 2026, doi: 10.1016/j.psep.2026.108830.

[23] K. A. Severson *et al.*, “Data-driven prediction of battery cycle life before capacity degradation,” *Nat. Energy*, vol. 4, no. 5, Art. no. 5, May 2019, doi: 10.1038/s41560-019-0356-8.
[24] T. Lin, S. Chen, S. J. Harris, T. Zhao, Y. Liu, and J. Wan, “Investigating explainable transfer learning for battery lifetime prediction under state transitions,” *eScience*, vol. 4, no. 5, p. 100280, Oct. 2024, doi: 10.1016/j.esci.2024.100280.
[25] N. Guo, S. Chen, J. Tao, Y. Liu, J. Wan, and X. Li, “Semi-supervised learning for explainable few-shot battery lifetime prediction,” *Joule*, vol. 8, no. 6, pp. 1820–1836, Jun. 2024, doi: 10.1016/j.joule.2024.02.020.
[26] “Status, challenges, and promises of data-driven battery lifetime prediction under cyber-physical system context - Liu - IET Cyber-Physical Systems: Theory & Applications - Wiley Online Library.” Accessed: Jun. 16, 2024. [Online]. Available: https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/cps2.12086
[27] O. S. Mahdy *et al.*, “Quantitative evaluation of thermal runaway in lithium-ion batteries under critical heating conditions to enhance safety,” *Sci. Rep.*, vol. 15, no. 1, p. 24004, Jul. 2025, doi: 10.1038/s41598-025-07824-7.
[28] K. A. A. Arafat, S. M. Yousuf Bhuiyan, R. Mahamud, and I. Parvez, “Investigating the Performance of Different Machine Learning Models for Forecasting Li-ion Battery Core Temperature Under Dynamic Loading Conditions,” in *2024 IEEE International Conference on Electro Information Technology (eIT)*, May 2024, pp. 1–7. doi: 10.1109/eIT60633.2024.10795487.
[29] Q. Lin, K. Fu, J. Ding, K. Sun, and P. Tan, “Advances in Thermal Modeling and Simulation of Lithium-Ion Batteries with Machine Learning Approaches,” *Adv. Intell. Discov.*, vol. n/a, no. n/a, p. e202500147, doi: 10.1002/aidi.202500147.
[30] Z. Liu and Y. Li, “Lithium Battery Thermal-Runaway Monitoring Based on Whole-Feature Neural Networks,” *J. Electrochem. Soc.*, vol. 171, no. 8, p. 080517, Sep. 2024, doi: 10.1149/1945-7111/ad69c5.